RUDRA BHALLA

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Summary

B.Tech Computer Science student skilled in Java, Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, and Machine Learning. Developed multiple software and ML projects using Java, Python, SQL, Scikit-learn, and TensorFlow. Strong analytical and problem-solving abilities with a keen interest in Software Engineering, distributed systems, and scalable application development.

Education

B.Tech in Computer Science Engineering 2023 – 2027
Uttranchal University
CGPA: 8.62

Class 12th, *PSEB* 2022
86.6

Class 10th, *CBSE* 2020
86

Skills

Programming Languages
- Java
- Python
- SQL

Core Computer Science
- Object-Oriented Programming (OOP)
- Database Management Systems (DBMS)
- Operating Systems
- Computer Networks

Machine Learning
- Machine Learning
- Deep Learning
- NumPy
- Pandas
- Scikit-learn
- TensorFlow
- Keras
- Matplotlib

Data Structures & Algorithms
- Arrays
- Linked Lists
- Stacks & Queues
- Trees & Graphs
- Hashing
- Recursion & Backtracking
- Dynamic Programming
- Binary Search

Web Development
- HTML
- CSS
- JavaScript

Tools & Technologies
- Git
- GitHub
- VS Code
- IntelliJ IDEA
- Jupyter Notebook
- Streamlit

Projects

Online Quiz Application, *Technologies: Java, OOP, Collections*

- Developed a Java-based quiz application supporting multiple-choice questions and automated score calculation.
- Implemented user-friendly interfaces and efficient data management for quiz administration.
- Applied Object-Oriented Programming principles to create a modular and maintainable application architecture.
- Generated real-time results and performance summaries for users.

Library Management System, *Technologies: Java, JDBC, MySQL, OOP*

- Developed a Library Management System to manage book issuance, returns, inventory tracking, and member records.
- Implemented CRUD operations using JDBC and MySQL for efficient database management.
- Designed a modular architecture using Object-Oriented Programming principles to improve maintainability.
- Added search and record management functionalities to streamline library operations.

Vehicle Image Classification System, *Technologies: Python, TensorFlow, Keras, CNN, NumPy, Pandas*

• Built a CNN-based image classification model to distinguish between cars and bikes.
• Performed image preprocessing, augmentation, and model training using TensorFlow and Keras.
• Evaluated model performance using accuracy metrics and confusion matrix analysis.
• Improved prediction accuracy through hyperparameter tuning and model optimization techniques.

Achievements

Achievement

• Solved 200+ Data Structures & Algorithms problems on LeetCode and other coding platforms.
• Completed a Machine Learning Internship, gaining hands-on experience in data preprocessing, model training, and evaluation.